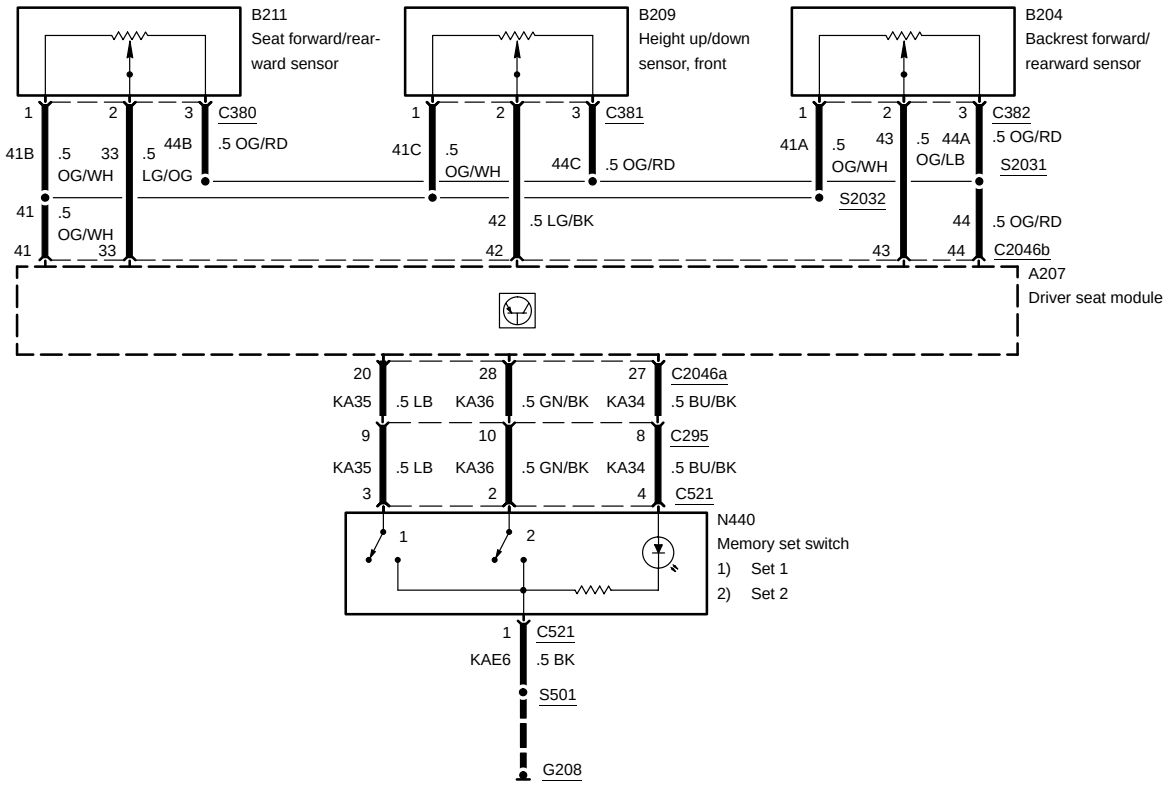with memory -4